

MTH203: Multivariate Calculus

Monsoon 2025

Name: _____

Roll No.: _____

Tutorial Quiz 3

Question:

1. If $w = x^2 + y^2 + z^2$ and $z = x^2 + y^2$, find $\left(\frac{\partial w}{\partial z}\right)_x$ **1**

2. If $w = x^2 + y - z + \sin(t)$ and $x + y = t$, find $\left(\frac{\partial w}{\partial z}\right)_{y,t}$ **1**

3. If $z = x + f(u)$, where $u = xy$, show that

$$x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = x$$

1

Please note: There is no partial marking.